



Deployment Guide — AWS Invoice Pipeline

This guide outlines the step-by-step instructions to deploy the event-driven invoice processing pipeline and React dashboard in your own AWS account.



Prerequisites

Ensure you have the following installed on your machine:

- **Node.js 18+** & **pnpm** (installed globally)
 - **AWS CLI v2** configured with your IAM credentials
 - **AWS SAM CLI** (Serverless Application Model)
 - **make** (utility for running build scripts)
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Step 1: AWS Console Preparations

Before launching the deployment pipeline, you must manually set up SES verification and Bedrock access in the AWS Console.

1. Verify SES Email Identities

SES requires verification before sending emails to or from an address (especially in sandbox mode).

1. Navigate to the **Amazon SES Console** → **Verified identities**.
2. Click **Create identity** and select **Email address**.
3. Enter your email (e.g., `mohad.asif0407@gmail.com`).
4. Click **Create identity**.
5. Check your inbox and click the **AWS Verification Link** inside the email.
6. If your account is in Sandbox mode, repeat these steps to verify the *recipient* email address if it is different from the sender.

2. Enable Amazon Bedrock Model Access

By default, model access to Anthropic Claude is not enabled.

1. Navigate to the **Amazon Bedrock Console**.
 2. Select your deployment region (e.g., **Mumbai ap-south-1** or **N. Virginia us-east-1**).
 3. In the left navigation pane, scroll to the bottom and click **Model access**.
 4. Click **Manage model access** (top-right).
 5. Check the box next to **Anthropic** → **Claude 3 Sonnet** and **Claude 3 Haiku**.
 6. Click **Save changes** (access is usually granted within 5 minutes).
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Step 2: Local Environment Setup

Configure your environment variables before deployment:

1. Copy the example `.env` file at the root:

```
cp .env.example .env
```

2. Fill in the keys following the instructions in [env_setup_guide.md](#). Ensure the following are set:

- `AWS_ACCOUNT_ID` : Your 12-digit account number.
- `SES_SENDER_EMAIL` & `REVIEWER_EMAIL` : The emails verified in Step 1.
- `AWS_REGION` : Set to `ap-south-1` (or your preferred region).

Step 3: Compile and Deploy Infrastructure

The root `Makefile` automates the entire process:

1. Install Dependencies:

```
make install
```

This downloads and builds dependencies for all Lambda modules and the frontend.

2. Build the SAM Template:

```
make build
```

This compiles TypeScript files and bundles Lambda functions via esbuild.

3. Deploy stack to AWS (First Time):

```
make deploy-guided
```

Provide the following parameters when prompted by the SAM CLI:

- **Stack Name:** `aws-invoice-pipeline`
- **AWS Region:** `ap-south-1` (or matching your `.env`)
- **Parameter SenderEmail:** Your verified SES email
- **Parameter ReviewerEmail:** Your verified reviewer email
- **Confirm changes before deploy:** `Yes`
- **Allow SAM CLI IAM role creation:** `Yes`
- **Save arguments to configuration file:** `Yes` (updates `samconfig.toml`)

4. Verify Outputs:

After successful deployment, copy the outputs from the terminal:

- `ApiEndpoint` : The API Gateway REST URL (e.g., `https://xxxx.execute-api.ap-south-1.amazonaws.com/prod`).
- `StateMachineArn` : Step Functions workflow ARN.

5. Update your root `.env` and [frontend/.env](#) with the `ApiEndpoint` value:

```
# In frontend/.env
VITE_API_BASE_URL=https://xxxx.execute-api.ap-south-1.amazonaws.com/prod
```

Step 4: Host React Frontend (AWS Amplify)

Use the AWS Amplify Console to host the React UI dashboard with auto-deployments linked to Git:

1. Push your repository to **GitHub / GitLab / Bitbucket**.

2. Navigate to the **AWS Amplify Console**.
3. Click **Create new app** → **Host web app**.
4. Authorize your Git provider and select the repository/branch (`main`).
5. Amplify will auto-detect the build specifications inside [amplify.yml](#) which we configured at the root.
6. In **Advanced Settings**, add the environment variables:
 - Set `VITE_API_BASE_URL` to your CloudFormation `ApiEndpoint` URL.
 - Set `VITE_AWS_REGION` to your AWS region code.
7. Click **Save and deploy**. Amplify will provision, build, and host the dashboard on a public `amplifyapp.com` subdomain!